

Efficacy and Safety of Upadacitinib or Elsubrutinib Alone or in Combination for Patients With Systemic Lupus Erythematosus: A Phase 2 Randomized Controlled Trial

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Objective. The 48-week, phase 2 SLEek study (NCT03978520) evaluated the efficacy and safety of upadacitinib (JAK inhibitor) and elsubrutinib (BTK inhibitor) alone or in combination (ABBV-599) in adults with moderately to severely active systemic lupus erythematosus (SLE).

Methods. Patients were randomized 1:1:1:1 to elsubrutinib 60 mg and upadacitinib 30 mg once daily (ABBV-599 high dose), elsubrutinib 60 mg and upadacitinib 15 mg once daily (ABBV-599 low dose), elsubrutinib 60 mg once daily (QD), upadacitinib 30 mg QD, or placebo QD. The primary endpoint was the proportion of patients achieving both Systemic Lupus Erythematosus Responder Index 4 (SRI-4) and glucocorticoid dose ≤ 10 mg QD at week 24. Additional assessments through week 48 included British Isles Lupus Assessment Group-Based Composite Lupus Assessment (BICLA) and Lupus Low Disease Activity State (LLDAS) responses, number of flares, time to first flare, and adverse events.

Results. The study enrolled 341 patients. The ABBV-599 low dose and elsubrutinib arms were discontinued after a planned interim analysis showed lack of efficacy (no safety concerns). More patients achieved the primary endpoint with upadacitinib (54.8%; $P = 0.028$) and ABBV-599 high dose (48.5%; $P = 0.081$) versus placebo (37.3%). SRI-4, BICLA, and LLDAS response rates were higher for both upadacitinib and ABBV-599 high dose versus placebo at weeks 24 and 48. Flares were reduced, and time to first flare through week 48 was substantially delayed with both upadacitinib and ABBV-599 high dose versus placebo. No new safety signals were observed beyond those previously reported for upadacitinib or elsubrutinib.

Conclusion. Upadacitinib 30 mg alone or in combination with elsubrutinib (ABBV-599 high dose) demonstrated significant improvements in SLE disease activity and reduced flares and were well tolerated through 48 weeks.

INTRODUCTION

Systemic lupus erythematosus (SLE) is a chronic autoimmune disease characterized by disordered innate and adaptive

immune responses with activation of a wide spectrum of soluble mediators, autoantibodies, and immune complexes.^{1,2} Unchecked tissue inflammation results in symptoms such as arthritis, rashes, serositis, nephritis, and progressive

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organ damage.¹ Most patients with SLE are women of reproductive age.^{1,3}

Current treatment guidelines recommend antimalarials and immunosuppressive medications, including biologic treatments, to manage SLE symptoms and prevent disease flares and irreversible organ damage; the 2023 EULAR recommendations for the management of SLE additionally emphasize minimizing glucocorticoids, which cause substantial long-term toxicity.^{4,5} However, SLE can still be challenging to treat, and safe and effective therapeutic options are currently limited.^{4,6} There is substantial heterogeneity of SLE, and response to treatment can vary widely from patient to patient, even among those with similar clinical manifestations or within the same individual over time.^{1,7} Despite currently available treatments, patients with SLE can develop progressive organ damage, and both the disease and its treatments contribute to impaired quality of life; significant comorbidities such as chronic kidney disease, cardiovascular disease, and infections; and shortened lifespan.^{8–11} Given the numerous negative health and functional impacts of the disease and limited effective therapeutic options, a continuing unmet need exists for additional targeted therapies to manage SLE signs and symptoms.^{2,12,13}

Upadacitinib is an oral selective JAK inhibitor.^{14,15} Upadacitinib has been studied and demonstrated efficacy and acceptable safety in patients with various immune-mediated conditions, including rheumatoid arthritis (RA), psoriatic arthritis, ankylosing spondylitis, nonradiographic axial spondyloarthritis, ulcerative colitis, Crohn's disease, and atopic dermatitis.¹⁶ Elsubrutinib is an oral selective BTK inhibitor developed for the treatment of autoimmune disease.¹⁷ As a BTK inhibitor, elsubrutinib is hypothesized to inhibit B cell activation and immune complex-driven activation of dendritic cells and neutrophils, whereas JAK inhibition via upadacitinib is expected to disrupt signaling through type I and type II interferon (IFN) receptors, T cell activation, and the signaling of several other cytokines implicated in SLE (eg, interleukin [IL]-2, IL-4, IL-6, IL-10, and IL-15).^{2,18} Therefore, the ABBV-599 (elsubrutinib/upadacitinib) combination has the potential for concurrent inhibition of broader pathophysiologic processes of SLE than either treatment alone. The SLEek phase 2 study sought to evaluate the efficacy and safety of upadacitinib, elsubrutinib, and ABBV-599 in adults with moderately to severely active SLE. Herein, we report 24-week and 48-week efficacy and safety results from the SLEek study.

PATIENTS AND METHODS

Study design, patients, and treatment. SLEek (NCT03978520) was a 48-week, phase 2, randomized, double-blind, placebo-controlled, global, multicenter study evaluating the efficacy and safety of upadacitinib and elsubrutinib when administered as monotherapy or as ABBV-599 combination therapy (Supplementary Figure 1). The study included adult patients

aged 18 to 65 years with a clinical diagnosis of moderately to severely active SLE at least 24 weeks before screening with serologic positivity (defined as antinuclear antibody titer $\geq 1:80$, positive anti-double-stranded DNA [dsDNA] IgG, or positive anti-Smith) and meeting either the revised criteria for classification of SLE according to the 1997 update of the 1982 American College of Rheumatology criteria¹⁹ or the 2012 Systemic Lupus Erythematosus International Collaborating Clinics classification criteria.²⁰ Eligibility criteria also included Systemic Lupus Erythematosus Disease Activity Index 2000 (SLEDAI-2K)²¹ score ≥ 6 at screening (clinical score ≥ 4); if four points of the required clinical score were for arthritis, the patient must have had at least three tender and swollen joints. Patients were also required to have a Physician's Global Assessment of Disease Activity score ≥ 1 during the screening period and be on stable background treatment for ≥ 30 days before baseline with antimalarial(s), glucocorticoid (≤ 20 mg prednisone or prednisone-equivalent dose daily), and/or a single immunosuppressant (azathioprine, mycophenolate, leflunomide, cyclosporine, tacrolimus, or methotrexate). Prior exposure to belimumab or anifrolumab was permitted (after a washout period of ≥ 3 months). Glucocorticoid dose adjustments were performed according to the treating physician's clinical discretion; investigators were encouraged to taper steroids to ≤ 10 mg prednisone or equivalent daily by week 16 in patients who were stable or improving through an electronic prompt. Patients were ineligible if they had triple-positive antiphospholipid antibodies (lupus anticoagulant, anti-beta 2 glycoprotein 1 antibody IgG or IgM, and anticardiolipin antibody IgG or IgM), severely active lupus nephritis, or severely active neuropsychiatric SLE. An IFN score for each patient was evaluated at screening based on a validated assay for type I IFN RNA signature; patients were categorized as IFN high or IFN low (cut point of ≥ -0.25 or < -0.25 , respectively).

Patients were randomized 1:1:1:1 to receive elsubrutinib 60 mg and upadacitinib 30 mg once daily (ABV-599 high dose), elsubrutinib 60 mg and upadacitinib 15 mg once daily (ABV-599 low dose), elsubrutinib 60 mg once daily (QD) monotherapy, upadacitinib 30 mg QD monotherapy, or placebo QD for 48 weeks. Randomization was performed using interactive response technology at the screening visit and was stratified by baseline glucocorticoid dose (≤ 10 vs > 10 mg), screening SLEDAI-2K score (< 10 vs ≥ 10),²¹ IFN score (high vs low vs not applicable), and baseline immunosuppressant use (yes vs no). The study was conducted in accordance with the protocol, International Council for Harmonisation guidelines, and applicable regulations, guidelines, and ethical principles originating from the Declaration of Helsinki. The study protocol was reviewed and approved by an institutional review board, and all patients provided written informed consent before screening or undergoing study-specific procedures.

A planned unblinded interim analysis was performed when 50% of the planned patients completed their week

24 assessments or withdrew from the study. This analysis was performed to reassess the treatment regimens in the study and evaluate the benefit/risk for continued evaluation. The interim analysis was performed by an independent team (the internal executive review committee) that was separate from the blinded study team, who remained blinded through the week 24 primary analysis. Study sites and patients remained blinded to treatment assignment throughout the study.

Assessments. *Efficacy assessments.* The primary endpoint was the proportion of patients achieving both Systemic Lupus Erythematosus Responder Index 4 (SRI-4) response²² and glucocorticoid dose ≤ 10 mg QD at week 24. Secondary endpoints were evaluated at week 24 and included the proportion of patients achieving SRI-4 response, British Isles Lupus Assessment Group-Based Composite Lupus Assessment (BICLA) response,²³ and Lupus Low Disease Activity State (LLDAS) response²⁴; SRI-4, BICLA, and LLDAS responses are defined in the Supplementary Methods. Other secondary endpoints included change from baseline to week 24 in glucocorticoid dose and the number of flare events per patient-year (PY) through week 24 (overall, mild/moderate, or severe) as measured by the Safety of Estrogens in Lupus Erythematosus National Assessment Systemic Lupus Erythematosus Disease Activity Index Flare Index (SFI).^{25,26}

Additional efficacy endpoints included the proportion of patients achieving SRI-4, BICLA, or LLDAS responses at week 48, change from baseline to week 48 in glucocorticoid dose, number of flare events per PY through week 48 (overall, mild/moderate, or severe) as measured by SFI, time from study drug initiation to first flare through week 48 as measured by SFI after study drug administration, the proportion of patients achieving 50% reduction of tender or swollen joints at week 48 among patients with at least six affected joints at baseline (Joint Count-50), and the proportion of patients achieving 50% reduction in Cutaneous Lupus Erythematosus Disease Area and Severity Index activity score at week 48 among patients with baseline Cutaneous Lupus Erythematosus Disease Area and Severity Index activity score ≥ 10 (CLASI-50).²⁷

Serum levels of anti-dsDNA IgG and complement components C3 and C4 were evaluated using enzyme-linked immunosorbent assay and turbidimetry, respectively (both from LabCorp). Change from baseline values by visit over time was calculated for levels of anti-dsDNA IgG (among those with detectable levels at screening) and complement components C3 and C4 (among those with decreased baseline levels). The IFN gene signature was assessed from whole blood by polymerase chain reaction using the validated four-gene IFN panel (*HERC5*, *RADS2*, *IFI27*, and *IFIT1*; DxTertity).

Pharmacokinetics and exposure-response analyses. Blood samples were collected at baseline and weeks 4, 12, 20, and 24 to measure the plasma concentrations of upadacitinib and

elsubrutinib. A population pharmacokinetic modeling approach was used to assess the pharmacokinetics of upadacitinib and elsubrutinib. Pharmacokinetic and efficacy data were used for exposure-response analyses, with logistic regression models used to characterize the relationship between the model-estimated average upadacitinib or elsubrutinib plasma concentrations over a dosing interval and achievement of key efficacy endpoints at weeks 24 and 48. Details are provided in the Supplementary Methods.

Safety assessments. Safety was assessed in all patients who received at least one dose of the study drug by monitoring treatment-emergent adverse events (TEAEs), defined as adverse events with an onset date that was on or after the first dose of the study drug, and no more than 30 days after the last dose of the study drug. TEAEs were summarized using Medical Dictionary for Regulatory Activities (version 25.0) system organ classes and preferred terms. Safety assessments also included monitoring of adverse events of special interest, TEAEs leading to study drug discontinuation, and TEAEs leading to death. Potential blinded major adverse cardiovascular events (MACEs) and venous thromboembolism (VTE) events were reviewed and adjudicated by an independent external committee.

Statistical analysis. The planned sample size was 325 patients (65 per cohort), based on the primary endpoint: assuming a placebo response rate of 25% at week 24, a two-sided χ^2 test with $\alpha = 10\%$ would provide 78% to 91% power to detect a difference of 20% to 25% between an active treatment group and the placebo group. Patient demographics, exposure-response analyses, and safety data were reported with descriptive statistics. For categorical efficacy endpoints, differences between treatment groups were assessed using the Cochran-Mantel-Haenszel test and stratified by the randomization stratification factors; patients who missed a prespecified study visit were categorized as nonresponders for that visit, except for those missing data due to COVID-19 infection or logistical restriction, which were handled by multiple imputation. Continuous endpoints were evaluated as change from baseline using a mixed-effect model for repeated measures. The Cox proportional-hazards model was used to assess the treatment effect for time to first flare based on observed data; patients who did not have a flare were censored at the date of the last nonmissing SFI measurement or at the time of discontinuation. The prespecified two-sided alpha level was 0.1; because there was no multiplicity control, all *P* values except for the primary endpoint were considered nominal. For post hoc biomarker analyses, the two-sided alpha level was 0.05.

RESULTS

Patients. The study enrolled 341 patients (ABBV-599 high dose, *n* = 68; upadacitinib 30 mg, *n* = 62; ABBV-599 low dose,

$n = 69$; elsubrutinib 60 mg, $n = 67$; placebo, $n = 75$) and was conducted between July 2019 and July 2022 (Supplementary Figure 2). There were 112 study sites in 20 countries across North and South America, Europe, Asia, and Australia. After a planned interim analysis when 50% of the patients reached week 24 or withdrew from the study, the ABBV-599 low dose and elsubrutinib 60 mg treatment arms were discontinued due to lack of efficacy (Supplementary Figure 3) with no safety concerns. Of 205 patients in the continued groups (ABBV-599 high dose, upadacitinib 30 mg, or placebo), baseline demographics and disease characteristics were comparable across treatment arms (Table 1); baseline demographics and disease characteristics of the discontinued groups are reported in Supplementary Table 1. Most patients were female (91%–100% across groups, consistent with the general population with SLE) and White (54%–66%). Mean duration of study drug exposure was 290 days for ABBV-599 high dose, 298 days for upadacitinib 30 mg, and 286 days for placebo.

Efficacy. The primary endpoint (proportion of patients achieving both SRI-4 response and glucocorticoid dose ≤ 10 mg QD at week 24) was met for both the groups that received ABBV-599 high dose and upadacitinib 30 mg versus placebo (response rates: ABBV-599 high dose, 48.5%, $n/N = 33/68$ [$P = 0.081$]; upadacitinib 30 mg, 54.8%, $n/N = 34/62$ [$P = 0.028$]; placebo, 37.3%, $n/N = 28/75$; Figure 1A). A higher proportion of patients achieved SRI-4, BICLA, and LLDAS responses at week 24 in both the groups that received ABBV-599 high dose and upadacitinib 30 mg versus placebo (all $P < 0.10$); these responses were maintained through week 48, at which time a higher proportion of patients achieved SRI-4, BICLA, and LLDAS responses while receiving ABBV-599 high dose and upadacitinib 30 mg versus placebo (all $P < 0.10$; Figure 1B). Incidence of overall flares as measured by SFI was lower in patients receiving both ABBV-599 high dose and upadacitinib 30 mg compared with those receiving placebo through week 24 ($P < 0.05$) and week 48 ($P < 0.01$; Figure 2). Mild/moderate flares were also less common in patients receiving ABBV-599 high dose and upadacitinib 30 mg versus placebo through weeks 24 ($P < 0.10$) and 48 ($P < 0.05$). Severe flares were also lower when receiving ABBV-599 high dose and upadacitinib 30 mg versus placebo through weeks 24 and 48, although the number of severe flares was small across all groups. Time to the first flare by SFI through week 48 was delayed in patients receiving ABBV-599 high dose and upadacitinib 30 mg versus placebo (hazard ratio, 0.591 [$P = 0.027$] and 0.476 [$P = 0.004$], respectively; Figure 2C).

Joint Count-50 response rates among patients with at least six affected joints at baseline were higher when they received ABBV-599 high dose ($P = 0.037$) and upadacitinib 30 mg ($P = 0.069$) versus placebo at week 48 (Figure 3A and B). CLASI-50 response rates among patients with baseline

Cutaneous Lupus Erythematosus Disease Area and Severity Index (CLASI) activity scores ≥ 10 were higher when they received ABBV-599 high dose versus placebo at week 24 ($P = 0.021$) and upadacitinib 30 mg versus placebo at week 48 ($P = 0.034$; Figure 3C and D), although the number of patients in each group with baseline CLASI activity scores ≥ 10 was low (ABBV-599 high dose, $n = 12$; upadacitinib, $n = 8$; placebo, $n = 14$). Small reductions in glucocorticoid doses were reported to a similar extent across treatment groups from baseline to weeks 24 and 48. At week 24, the least squares mean ($\pm$ SE) change from baseline in glucocorticoid dose was -0.5 (± 0.5) mg for patients receiving ABBV-599 high dose, -0.6 (± 0.5) mg for those receiving upadacitinib 30 mg, and -0.7 (± 0.5) mg for those receiving placebo. At week 48, the least squares mean ($\pm$ SE) change from baseline in glucocorticoid dose was -1.5 (± 0.5) mg for patients receiving ABBV-599 high dose, -1.2 (± 0.5) mg for those receiving upadacitinib 30 mg, and -1.5 (± 0.5) mg for those receiving placebo.

In patients with detectable anti-dsDNA autoantibody levels at screening, anti-dsDNA levels were substantially decreased in those receiving ABBV-599 high dose and upadacitinib 30 mg versus placebo at week 48 (Figure 4A). Among patients with low levels of complement C3 or C4 at baseline, treatment with ABBV-599 high dose and upadacitinib 30 mg was associated with further reduction of C3 or C4 levels at week 2 compared with baseline (C3, $P = 0.028$ and $P = 0.007$; C4, $P = 0.093$ and $P = 0.183$, respectively), but no significant change was observed for C3 and C4 levels at week 48 (Figure 4B and C). There was a decrease in the IFN gene signature module at week 4 that was maintained at week 24 posttreatment in patients receiving ABBV-599 high dose and upadacitinib 30 mg compared with baseline ($P < 0.0001$ at weeks 4 and 24; Figure 4D). The discontinued treatments (in patients receiving ABBV-599 low dose and elsubrutinib 60 mg) had no significant impact on these biomarkers at week 24 or week 48 (Supplementary Figure 4).

Pharmacokinetics and exposure-response analyses.

Plasma exposures following administration of upadacitinib 15 or 30 mg and elsubrutinib 60 mg were comparable to those previously observed in patients with RA (Supplementary Table 2). No clear or consistent exposure-response trends were observed when patients received elsubrutinib (Supplementary Figure 5). Higher efficacy responses were observed with increasing exposures of upadacitinib across all key efficacy endpoints evaluated at weeks 24 (Supplementary Figure 6) and 48 (Supplementary Figures 7 and 8A). The addition of elsubrutinib plasma concentration as a predictor of efficacy did not result in any meaningful improvement in upadacitinib exposure-response models in patients with SLE. Simulations based on logistic regression models developed for the efficacy endpoints demonstrated that upadacitinib 30 mg is predicted to provide approximately 8% to 16% higher response rates compared with upadacitinib 15 mg across all efficacy endpoints at week 48 (Supplementary

Table 1. Baseline demographics and characteristics of the participants*

Parameter	ABBV-599 high dose (n = 68)	Upadacitinib 30 mg (n = 62)	Placebo (n = 75)
Female, n (%)	62 (91.2)	57 (91.9)	75 (100)
Race, n (%)			
Asian	14 (20.6)	13 (21.0)	23 (30.7)
Black	4 (5.9)	9 (14.5)	4 (5.3)
Multiple	2 (2.9)	6 (9.7)	2 (2.7)
Other ^a	3 (4.4)	0	3 (4.0)
White	45 (66.2)	34 (54.8)	43 (57.3)
Age, mean (±SD), y	42.7 (±11.3)	42.5 (±11.9)	41.7 (±12.1)
BMI, mean (±SD), kg/m ²	26.4 (±7.0)	27.4 (±7.4)	27.2 (±7.5)
Time since onset of lupus, mean (±SD), y	9.6 (±7.5)	11.0 (±6.9)	9.4 (±7.3)
Glucocorticoid use, n (%)	52 (76.5)	42 (67.7)	56 (74.7)
≤10 mg/d ^b	40 (76.9)	31 (73.8)	40 (71.4)
>10 mg/d ^b	12 (23.1)	11 (26.2)	16 (28.6)
Daily glucocorticoid dose, mg/day, mean (±SD)	6.7 (±6.4)	6.2 (±6.1)	7.9 (±7.1)
Antimalarial use, n (%)	62 (91.2)	51 (82.3)	67 (89.3)
Immunosuppressant use, n (%)	26 (38.2)	25 (40.3)	28 (37.3)
Mycophenolate	9 (13.2)	10 (16.1)	14 (18.7)
Azathioprine	13 (19.1)	11 (17.7)	9 (12.0)
Methotrexate	10 (14.7)	7 (11.3)	16 (21.3)
Calcineurin inhibitor ^c	4 (5.9)	1 (1.6)	4 (5.3)
Leflunomide	0	3 (4.8)	1 (1.3)
SLEDAI-2K status, n (%)			
<10	40 (58.8)	35 (56.5)	45 (60.0)
≥10	28 (41.2)	27 (43.5)	30 (40.0)
SLEDAI-2K score, mean (±SD)	8.9 (±2.8)	9.0 (±2.8)	9.1 (±3.9)
Among patients with score <10	7.1 (1.1)	7.3 (1.2)	6.7 (1.7)
Among patients with score ≥10	11.5 (2.3)	11.4 (2.6)	12.7 (3.4)
BILAG A or B organ system involvement, n (%)			
Constitutional	1 (1.5)	1 (1.6)	1 (1.3)
Mucocutaneous	34 (50.0)	30 (48.4)	35 (46.7)
Neuropsychiatric	0	0	0
Musculoskeletal	57 (83.8)	52 (83.9)	59 (78.7)
Cardiorespiratory	1 (1.5)	3 (4.8)	1 (1.3)
Gastrointestinal	0	1 (1.6)	0
Ophthalmic	0	0	0
Renal	2 (2.9)	0	4 (5.3)
Hematologic	1 (1.5)	0	0
PhGA DAS, mean (±SD)	1.8 (±0.4)	1.7 (±0.4)	1.8 (±0.4)

* ABBV-599 high dose, elsubrutinib 60 mg and upadacitinib 30 mg once daily. BILAG, British Isles Lupus Assessment Group; BMI, body mass index; PhGA DAS, Physician's Global Assessment of Disease Activity Score; SLEDAI-2K, Systemic Lupus Erythematosus Disease Activity Index 2000.

^a Other includes American Indian, Alaska native, Native Hawaiian, other Pacific islander, or other race.

^b Among patients who were using glucocorticoids at baseline.

^c This includes tacrolimus and cyclosporine.

Figure 8B). No significant effect was observed for baseline immunosuppressant use, baseline IFN score, demographic characteristics, or baseline disease duration on upadacitinib exposure-response relationships across all endpoints evaluated.

Safety. Adverse events are reported through week 48 for the continued groups in Table 2 and for all treatment groups through week 24 in Supplementary Table 3. The incidence of serious TEAEs through week 48 was lower in patients receiving ABBV-599 high dose (10.3%) than in those receiving upadacitinib 30 mg (21.0%) or placebo (17.3%). There were no patients with malignancy or adjudicated VTE. There were three nonfatal adjudicated MACEs, including one myocardial infarction (placebo) and two nonfatal strokes due to ruptured cerebral aneurysms (one in

a patient receiving ABBV-599 high dose and one in a patient receiving upadacitinib 30 mg); all events were assessed by the investigators as having no reasonable possibility of being related to study drug. There was a higher incidence of serious infections through week 48 in the groups that received ABBV-599 high dose and upadacitinib 30 mg groups than in the group that received placebo (ABBV-599 high dose, 7.4%; upadacitinib, 11.3%; placebo, 4.0%); these patients had underlying risk factors for serious infections including the use of concomitant immunosuppressive medications. There was a numerically higher incidence of herpes zoster infections through week 48 in the groups that received ABBV-599 high dose and upadacitinib 30 mg compared with those who received placebo (ABBV-599 high dose, 11.8%; upadacitinib, 6.5%; placebo, 4.0%). There were three

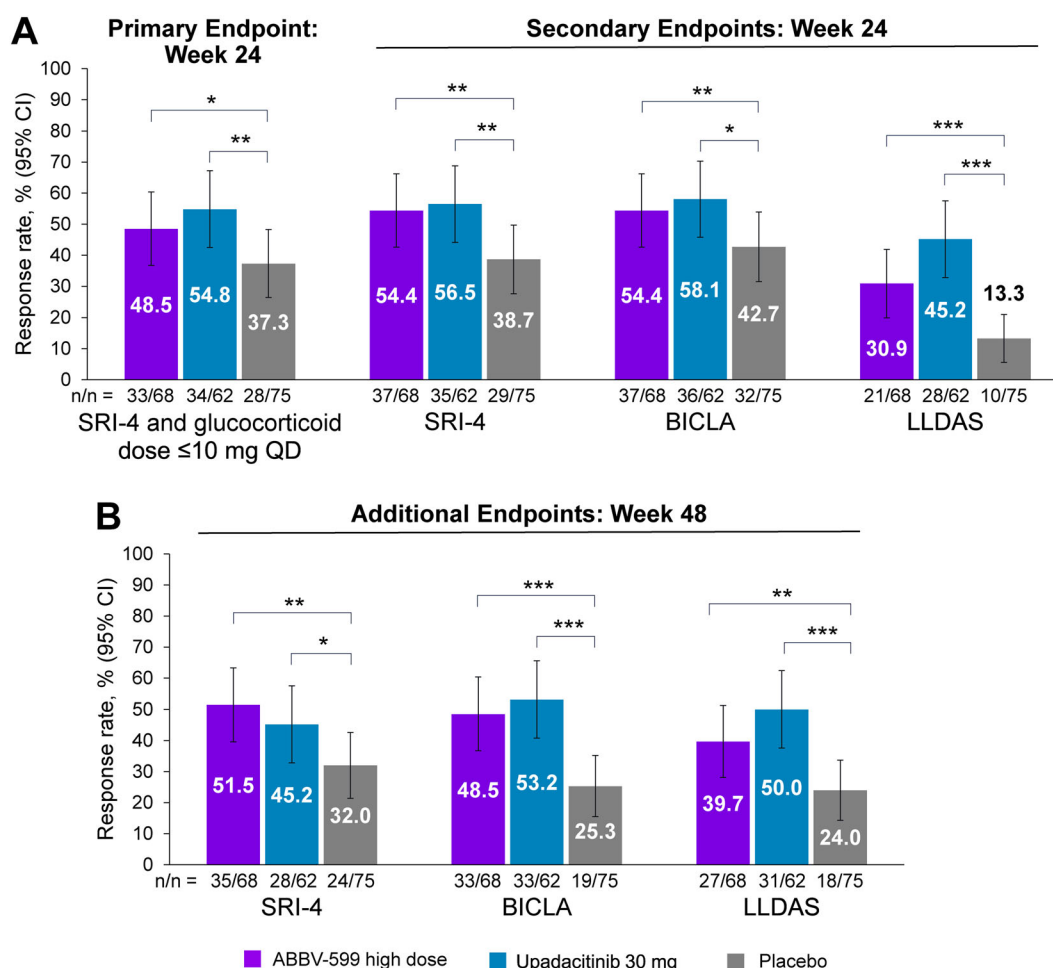


Figure 1. Key efficacy endpoints at (A) week 24 and (B) week 48. * $P < 0.10$, ** $P < 0.05$, *** $P < 0.01$ versus placebo. ABBV-599 high dose, elsubrutinib 60 mg and upadacitinib 30 mg once daily; BICLA, British Isles Lupus Assessment Group-Based Combined Lupus Assessment; CI, confidence interval; LLDAS, Lupus Low Disease Activity State; QD, once daily; SRI-4, Systemic Lupus Erythematosus Responder Index 4.

TEAEs resulting in death, two of which were in the group that discontinued elsubrutinib monotherapy. None of the deaths were considered possibly related to the study drug according to the investigators; two deaths were related to COVID-19 (ABBV-599 high dose, $n = 1$; elsubrutinib, $n = 1$), and one was an accidental death due to toxic effects of prescribed pain medications and ethanol (elsubrutinib).

DISCUSSION

The phase 2, multicenter, double-blind, placebo-controlled SLEek study examined the efficacy and safety of upadacitinib and elsubrutinib used as single agents and in combination (ABBV-599) for the treatment of patients with SLE. After a planned interim analysis, two dosing groups were discontinued due to lack of efficacy but not due to safety concerns; these groups included patients receiving elsubrutinib 60 mg QD as monotherapy or in combination with upadacitinib 15 mg QD (ABBV-599 low dose). The two remaining active treatment

groups, receiving upadacitinib 30 mg QD as monotherapy and combined with elsubrutinib 60 mg QD (ABBV-599 high dose), both met the primary endpoint (SRI-4 response with glucocorticoid dose ≤10 mg per day) and multiple secondary endpoints, demonstrating significant improvements in SLE disease activity, attainment of a low disease activity state, reduction in disease flares, and delayed time to first flare compared with placebo. Efficacy was sustained through 48 weeks of evaluation. In this study, the addition of elsubrutinib to upadacitinib did not substantially increase the proportion of treatment responders for any efficacy endpoints compared with patients receiving upadacitinib alone despite achieving plasma exposures associated with adequate target occupancy during the dosing interval.²⁸ Findings from the current study are consistent with previously reported lack of efficacy with BTK inhibitors as monotherapies for patients with SLE^{29,30} and consistent with a prior study in which patients received elsubrutinib in combination with upadacitinib to treat RA; the efficacy results in patients with RA were also driven by upadacitinib alone.²⁸

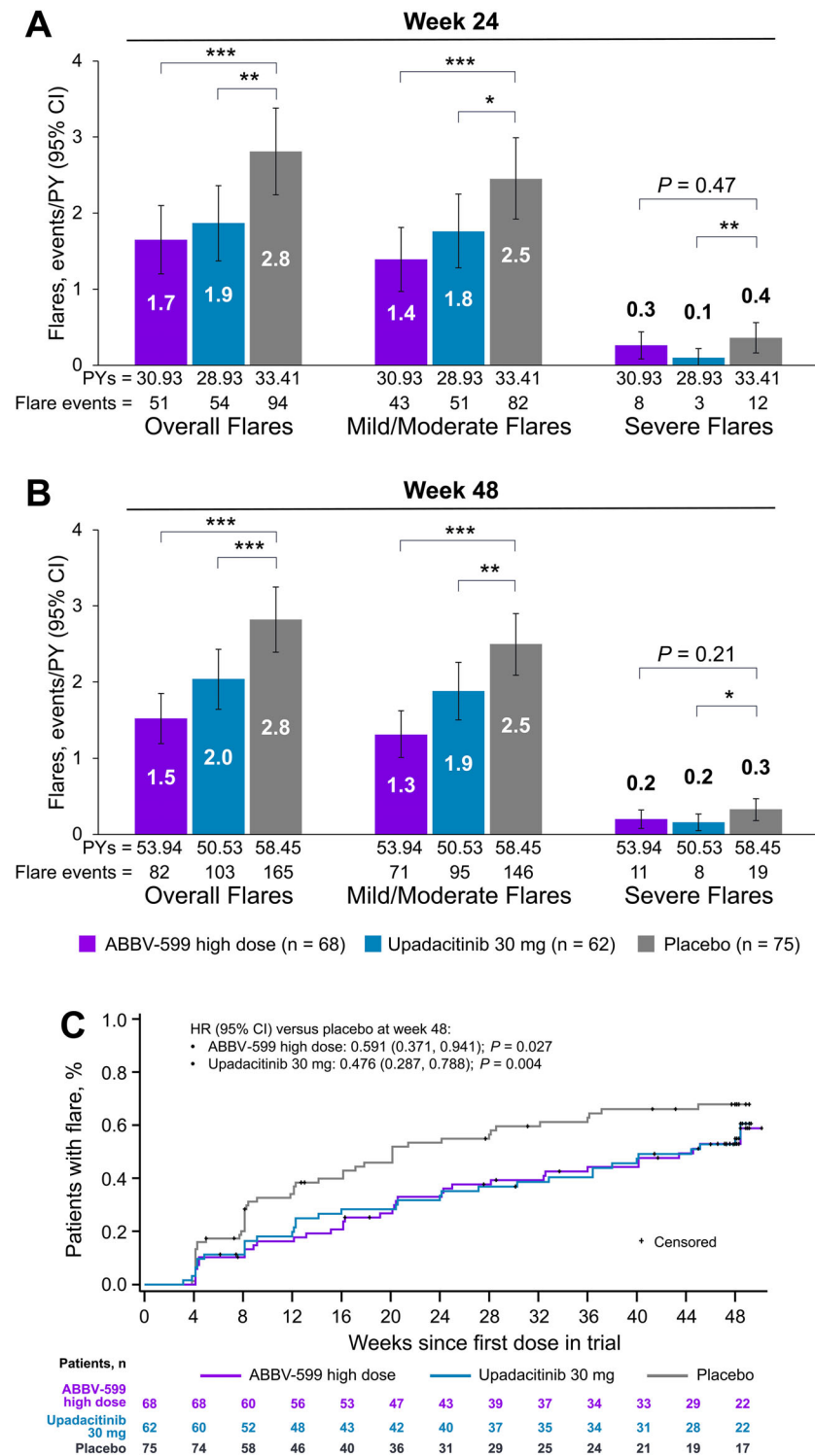


Figure 2. Number of flares by SFI through (A) week 24 and (B) week 48 and (C) time to first flare by SFI through week 48. * $P < 0.10$, ** $P < 0.05$, *** $P < 0.01$, versus placebo. ABBV-599 high dose, elsubrutinib 60 mg and upadacitinib 30 mg once daily; CI, confidence interval; HR, hazard ratio; PY, patient-year; SFI, Safety of Estrogens in Lupus Erythematosus National Assessment Systemic Lupus Erythematosus Disease Activity Index Flare Index; UPA, upadacitinib.

Based on the SLEek study, in which efficacy was not observed when a lower upadacitinib dose (15 mg) was combined with elsubrutinib (ABBV-599 low dose) and exposure-response

analyses demonstrated higher efficacy response rates with increased upadacitinib plasma exposures, 30 mg is predicted to be an efficacious upadacitinib dose for the treatment of patients with

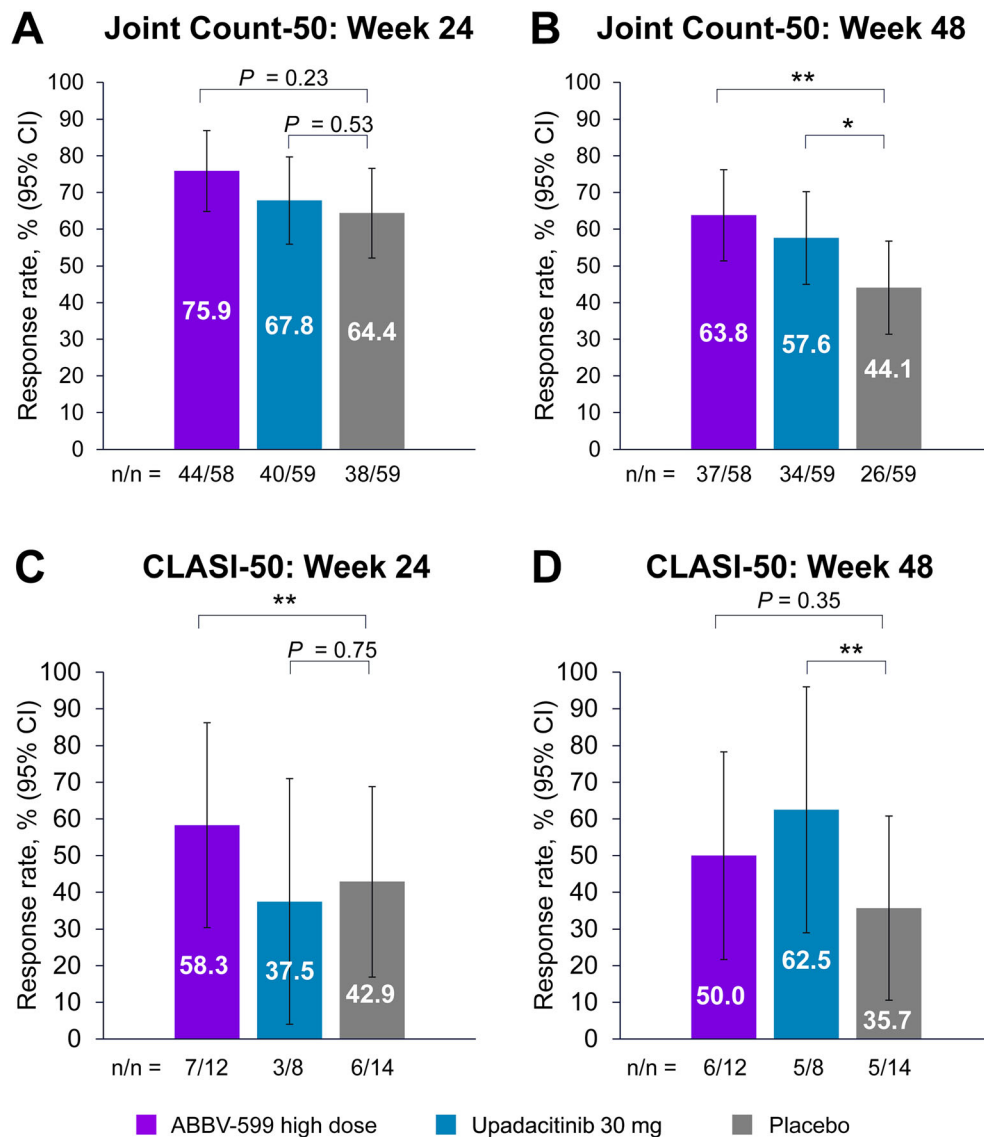


Figure 3. The proportion of patients achieving Joint Count-50 at (A) week 24 and (B) week 48, and the proportion of patients achieving CLASI-50 at (C) week 24 and (D) week 48. * $P < 0.10$, ** $P < 0.05$, versus placebo. ABBV-599 high dose, elsubrutinib 60 mg and upadacitinib 30 mg once daily; CI, confidence interval; CLASI-50, 50% reduction in Cutaneous Lupus Erythematosus Disease Area and Severity Index score among patients with baseline Cutaneous Lupus Erythematosus Disease Area and Severity Index score ≥ 10 ; Joint Count-50, 50% reduction of tender or swollen joints among patients with at least six affected joints.

SLE with moderate to severe disease activity. SLE has a distinct pathology and greater inflammation in multiple organ systems compared with RA and axial spondyloarthritis³¹; therefore, patients with SLE may require higher doses of upadacitinib, similar to those with other inflammatory diseases for which higher doses of upadacitinib have been approved, including ulcerative colitis, Crohn's disease, and atopic dermatitis.^{16,32–34} Further studies are needed to determine whether patients who achieve low disease activity may reduce the upadacitinib dose while maintaining disease control.

Upadacitinib pharmacokinetics in patients with SLE were consistent with the previously characterized pharmacokinetics in patients with RA,^{35–37} supporting the idea that the difference in optimal dose requirement between patients with SLE and RA

may be due to the difference in disease characteristics rather than differences in upadacitinib plasma concentrations. Exposure–response analyses for key efficacy endpoints demonstrated a clear relationship between increasing upadacitinib exposures in patients with SLE and efficacy and supported the greater efficacy benefit for 30 mg QD compared with 15 mg QD. The exposure–response relationships were consistent among different patient subgroups (eg, baseline immunosuppressant use, baseline IFN score); however, further assessment of the effect of these covariates on upadacitinib exposure–response relationship is warranted using data from larger clinical trials. Treatment with upadacitinib 30 mg or ABBV-599 high dose resulted in a decrease in anti-dsDNA and suppression of the type 1 IFN gene

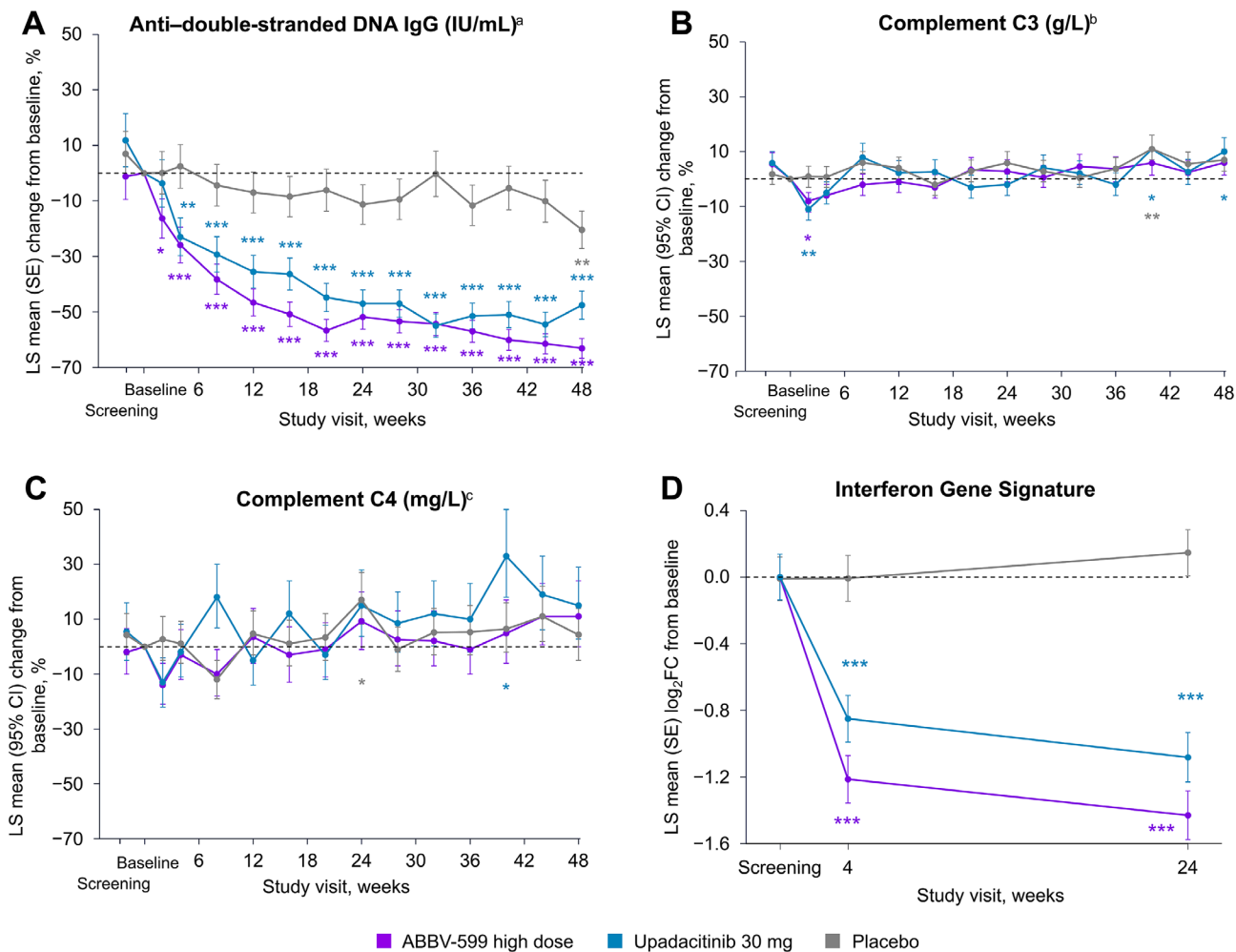


Figure 4. Assessment of change from baseline in (A) anti-double-stranded DNA autoantibody levels, complement (B) C3 and (C) C4 levels, and (D) interferon gene signature. $^*P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$ vs baseline value. ^aData represent patients with anti-double-stranded DNA autoantibody levels above the limit of quantification at SCR (ABBV-599 high dose, $n = 34$; upadacitinib 30 mg, $n = 33$; placebo, $n = 43$). ^bAmong patients with low baseline complement C3 levels (ABBV-599 high dose, $n = 27$; upadacitinib 30 mg, $n = 24$; placebo, $n = 34$). ^cAmong patients with low baseline complement C4 levels (ABBV-599 high dose, $n = 15$; upadacitinib 30 mg, $n = 12$; placebo, $n = 20$). ABBV-599 high dose, elsubrutinib 60 mg and upadacitinib 30 mg once daily; CI, confidence interval; $\log_2FC$, $\log_2$ fold change; LS, least squares.

signature, two hallmarks of SLE pathology. In patients with low baseline C3 or C4 levels, there was an initial decrease at week 2, followed by progressive normalization. This is hypothesized to be due to a known effect of IL-6 inhibition on complement production (rather than complement consumption)³⁸; in future studies, complement activation products will be measured to test this hypothesis.

No new safety signals were observed beyond previously reported data for upadacitinib³⁹ or elsubrutinib monotherapy.²⁸ There were numerically higher serious infections and herpes zoster in the groups that received ABBV-599 high dose and upadacitinib 30 mg, which was consistent with the known safety profile of JAK inhibitors and within the expected range. There were no VTE events observed despite the higher risk of VTE in patients with SLE.⁴⁰ Of note, although this study excluded patients with triple-positive antiphospholipid antibodies (Supplementary Methods),

patients with antiphospholipid syndrome, including those with prior VTE and/or on anticoagulation, were permitted in the study. There was no imbalance in MACEs across the placebo and treatment arms, and there were no malignancies. Safety will be further assessed in phase 3 studies, including more patients observed for a longer duration.

Anifrolumab, an approved treatment for patients with SLE,^{41,42} and deucravacitinib, an investigational agent with favorable results in a phase 2 trial,⁴³ each have some overlap with the mechanisms of upadacitinib. Anifrolumab targets the IFN $\alpha\beta$ receptor (IFNAR), which transmits type I IFN signals to a variety of cell types.^{41,42} Deucravacitinib inhibits the intracellular signal transducer tyrosine kinase 2, which receives signals through IFNAR, and has wider effects on IL-12 and IL-23 receptors.⁴³ Upadacitinib is a JAK inhibitor that was engineered for increased selectivity for JAK1,^{14,15} another key mediator of IFNAR signals.

Table 2. Safety overview through week 48*

Parameter, n (%)	ABBV-599 high dose (n = 68)	Upadacitinib 30 mg (n = 62)	Placebo (n = 75)
Safety overview			
Any TEAE	59 (86.8)	51 (82.3)	59 (78.7)
COVID-19-related TEAE	6 (8.8)	8 (12.9)	9 (12.0)
TEAE considered possibly related to study drug	29 (42.6)	20 (32.3)	25 (33.3)
TEAE leading to discontinuation of study drug	9 (13.2)	6 (9.7)	5 (6.7)
Serious TEAE	7 (10.3)	13 (21.0)	13 (17.3)
Death	1 (1.5) ^a	0	0
AESIs			
Serious infection	5 (7.4)	7 (11.3)	3 (4.0)
Opportunistic infection excluding TB and HZ ^b	0	1 (1.6)	1 (1.3)
HZ	8 (11.8)	4 (6.5)	3 (4.0)
Active TB	0	0	1 (1.3)
Possible malignancy	0	0	0
Malignancy	0	0	0
Anemia	3 (4.4)	2 (3.2)	3 (4.0)
Neutropenia	1 (1.5)	2 (3.2)	1 (1.3)
Lymphopenia	0	2 (3.2)	0
Renal dysfunction	1 (1.5)	1 (1.6)	0
Hepatic disorder	3 (4.4)	1 (1.6)	1 (1.3)
Adjudicated VTE	0	0	0
Any adjudicated MACE	1 (1.5)	1 (1.6)	1 (1.3)
Nonfatal MI ^c	0	0	1 (1.3)
Nonfatal stroke ^d	1 (1.5)	1 (1.6)	0

* ABBV-599 high dose, elsubrutinib 60 mg and upadacitinib 30 mg once daily. AESI, adverse event of special interest; HZ, herpes zoster; MACE, major adverse cardiovascular event; MI, myocardial infarction; TB, tuberculosis; TEAE, treatment-emergent adverse event; VTE, venous thromboembolism.

^a Death from acute respiratory failure due to COVID-19 before availability of COVID-19 vaccination; risk factors for severe COVID-19 included asthma, hypertension, and obesity. Death was assessed by the investigators as having no reasonable possibility of being related to the study drug.

^b Opportunistic infections were oropharyngeal candidiasis (placebo) and cytomegalovirus infection/*Pneumocystis jirovecii* pneumonia (one patient in the group that received upadacitinib 30 mg experienced both infections).

^c This occurred in a 61-year-old woman with a family history of cardiovascular disease and was assessed by the investigators as having no reasonable possibility of being related to study drug.

^d Nonfatal strokes were both due to ruptured cerebral aneurysms (one in a 47-year-old woman receiving ABBV-599 high dose and one in a 53-year-old woman receiving upadacitinib 30 mg); both assessed by the investigators as having no reasonable possibility of being related to the study drug.

JAK1 also modulates several other relevant SLE pathways including type II IFN, IL-2, IL-4, IL-6, IL-10, and IL-15. It is not yet known which targets of upadacitinib contribute to efficacy in individuals with SLE; for example, the role of inhibition of IL-2 signaling in patients with SLE is not well understood.⁴⁴ Another JAK inhibitor, baricitinib, which targets JAK1 and JAK2 signal transduction, completed phase 3 trials for SLE with mixed results; the primary endpoint was met for the SLE-BRAVE-1 study, but not the SLE-BRAVE-2 study.^{45,46} In both the SLE-BRAVE-1 and SLE-BRAVE-2 studies, there were high placebo response rates compared with other recent SLE trials, which the authors suggest may be in part due to the lack of an enforced glucocorticoid taper.^{45,46}

The complexity of SLE impedes clarity in accurately scoring and interpreting the equally complicated SLE outcome measures, which sometimes fall short in discriminating between minor and significant disease activity.^{47–49} Additionally, placebo-controlled trials cannot be conducted in the absence of background treatments, which in turn inflate placebo group responses.^{47,48} Given restrictions on the proportion of immunologically heterogeneous patients who could potentially respond to even the most promising targeted agent, coupled with limitations imposed by high

placebo responses and the noise of competing background treatments, many SLE trial outcomes over the past several decades have been difficult to interpret.⁵⁰ More recently, a subset analysis of earlier, disappointing SLE trials supported innovations to address high placebo response rates.⁴⁷ Many programs now include mandatory tapering of glucocorticoids and blinded review of data by clinicians with expertise in lupus to ensure sufficient disease activity at baseline, accurate diagnosis of SLE, and correct scoring of disease activity throughout the trial. The current trial employed blinded data review, which contributed to a placebo rate low enough for allowing detection of substantial treatment effects for upadacitinib in multiple endpoints.

A limitation of this study was that glucocorticoid tapering was encouraged but not mandatory, which may have contributed to lack of differences in glucocorticoid reduction between the active intervention arms versus placebo. On the other hand, baseline glucocorticoid doses were low in this population, potentially reducing the overall impact that background glucocorticoids might have had on study outcomes. Other limitations of the current phase 2 study include a modest sample size and data extending to only 48 weeks of evaluation, which limits the safety

database that is available to date. The main finding of this trial in patients with SLE is that daily oral upadacitinib 30 mg and ABBV-599 high dose met the primary endpoint (SRI-4 with a glucocorticoid dose ≤ 10 mg per day) and were effective in multiple outcome measures including disease activity, flares, time to first flare, and joint counts with an acceptable safety profile. A global phase 3 program to expand the evaluation of upadacitinib in patients with SLE has been initiated (NCT05843643).

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AUTHOR CONTRIBUTIONS

All authors were involved in drafting the article or revising it critically for important intellectual content, and all authors approved the final version to be published. All authors had full access to all the data in the study and take responsibility for the integrity of the data and the accuracy of the data analysis.

Study conception and design. Merrill, Tanaka, Doan, Friedman.

Acquisition of data. Friedman.

Analysis and interpretation of data. Merrill, Tanaka, D'Cruz, Vila-Rivera, Siri, Zeng, Saxena, Aringer, D'Silva, Cheng, Mohamed, Siovitz, Bhatnagar, Gaudreau, Doan, Friedman.

ROLE OF THE STUDY SPONSOR

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